

PHYS 143 – Mechanics and Thermal Physics
Week 8: Rotation – Supplementary Notes

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These notes support the Week 8 lecture. They contain background that was moved out of the contact time and are not needed for the lecture problems. Sections marked *extension* are not assessed.

1 Radians, rpm and units

An angle in radians is $\theta = s/r$, a ratio of two lengths, so it is dimensionless and the unit “rad” may be dropped or added to show which angle unit is meant. This is why $v = r\omega$ has units of m/s when ω is in rad/s. The same formula with ω in rpm or degrees per second gives a wrong number, because the factor $2\pi/60$ or $\pi/180$ is missing. Conversions: 1 rev = 2π rad; 1 rpm = 0.105 rad/s; 1 rad = 57.3° . Torque and work both have the unit $\text{N m} = \text{kg m}^2/\text{s}^2$, but torque is written as N m and energy as J: the torque is a vector along the axis, work is a scalar. The unit of I is kg m^2 and of angular momentum $\text{kg m}^2/\text{s}$.

2 Where $K = \frac{1}{2}I\omega^2$ comes from

Split the body into small pieces, piece i of mass m_i at perpendicular distance r_i from the fixed axis. All pieces have the same ω , so the speed of piece i is $v_i = r_i\omega$ and

$$K = \sum_i \frac{1}{2}m_i v_i^2 = \frac{1}{2} \left(\sum_i m_i r_i^2 \right) \omega^2 = \frac{1}{2}I\omega^2.$$

The same sum gives the law $\sum \tau = I\alpha$: the tangential force on piece i is $m_i r_i \alpha$, its torque is $m_i r_i^2 \alpha$, and the internal forces cancel in pairs (action and reaction along the line joining two pieces have equal and opposite torques about any axis).

3 The centre of mass

For particles the position of the centre of mass is $\vec{r}_{\text{cm}} = \sum m_i \vec{r}_i / M$. For a continuous body, $\vec{r}_{\text{cm}} = \frac{1}{M} \int \vec{r} dm$; a uniform body of symmetric shape has its centre of mass at the centre of symmetry. Adding Newton’s second law over all the pieces, the internal forces cancel and $\sum \vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}}$: the centre of mass moves as if all the mass were there and all the external forces acted on it. In the rod released from the horizontal (Contact 2) this gives the hinge force, $F_{\text{hinge}} = Mg - Ma_{\text{cm}} = Mg - M\alpha L/2 = \frac{1}{4}Mg$ at the instant of release, with $\alpha = 3g/2L$.

4 Proof of the parallel-axis theorem

Let the axis through the centre of mass be along z , and a parallel axis pass through the point $(d, 0)$ in the xy plane. For piece i at (x_i, y_i) measured from the centre of mass, the squared distance to the parallel axis is $(x_i - d)^2 + y_i^2 = r_i^2 - 2dx_i + d^2$. Then

$$I = \sum m_i (r_i^2 - 2dx_i + d^2) = I_{\text{cm}} - 2d \sum m_i x_i + Md^2.$$

Since the coordinates are measured from the centre of mass, $\sum m_i x_i = 0$, so $I = I_{\text{cm}} + Md^2$. Because $Md^2 \geq 0$, the smallest I among parallel axes is the one through the centre of mass.

5 Extension: I of a rod and a disc by integration

Thin uniform rod, mass M , length L , axis through the centre perpendicular to the rod. Take $dm = (M/L)dx$:

$$I = \int_{-L/2}^{L/2} x^2 \frac{M}{L} dx = \frac{M}{L} \cdot \frac{2}{3} \left(\frac{L}{2} \right)^3 = \frac{1}{12} ML^2.$$

The parallel-axis theorem with $d = L/2$ gives the value about one end: $\frac{1}{12} ML^2 + M(L/2)^2 = \frac{1}{3} ML^2$.

Solid uniform disc, mass M , radius R , axis through the centre perpendicular to the disc. Use thin rings of radius r and width dr , with mass $dm = \frac{M}{\pi R^2} 2\pi r dr$:

$$I = \int_0^R r^2 \frac{2M}{R^2} r dr = \frac{2M}{R^2} \frac{R^4}{4} = \frac{1}{2} MR^2.$$

6 Extension: the cross product in components

For vectors in the xy plane, $\vec{r} = (x, y)$ and $\vec{F} = (F_x, F_y)$, the torque about the origin along z is

$$\tau_z = xF_y - yF_x.$$

Example: $\vec{F} = (3.0, 4.0)$ N at $\vec{r} = (2.0, 1.0)$ m: $\tau_z = (2.0)(4.0) - (1.0)(3.0) = +5.0$ N m (anticlockwise). Check with $\tau = rF \sin \varphi$: $r = 2.24$ m, $F = 5.0$ N, and the angle between the vectors is $\arctan(4/3) - \arctan(1/2) = 53.1^\circ - 26.6^\circ = 26.6^\circ$, so $\tau = (2.24)(5.0) \sin 26.6^\circ = 5.0$ N m. ✓

7 Rolling without slipping: the role of friction

For a body of mass M and $I = cMR^2$ rolling down an incline of angle θ , take the centre of mass: $Mg \sin \theta - f = Ma$. About the centre, the friction force gives the torque $fR = I\alpha = cMR^2(a/R)$, so $f = cMa$. Then

$$a = \frac{g \sin \theta}{1 + c} \quad f = \frac{c}{1 + c} Mg \sin \theta.$$

The body rolls without slipping only if $f \leq \mu_s N = \mu_s Mg \cos \theta$, that is, $\mu_s \geq \frac{c}{1 + c} \tan \theta$. This static friction does no work, because the contact point is momentarily at rest, which is why the energy method of Contact 3 works. Check: for the hoop of Tutorial B6, $a = \frac{1}{2}(9.80)(0.5) = 2.45$ m/s² and $v^2 = 2as = 14.7$ m²/s², the same as gh from energy.

8 The linear and rotational quantities side by side

Quantity	Linear	Rotational (fixed axis)
position, velocity, acceleration	x, v, a	θ, ω, α
inertia	m	$I = \sum mr^2$
cause	F	$\tau = rF \sin \varphi$
Newton's second law	$\sum F = ma$	$\sum \tau = I\alpha$
kinetic energy	$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$
work, power	Fx, Fv	$\tau\theta, \tau\omega$
momentum	$p = mv$	$L = I\omega$
law with momentum	$\sum F = dp/dt$	$\sum \tau = dL/dt$

The analogy is exact for rotation about a fixed axis. Two differences matter: I depends on the chosen axis whereas m does not, and a torque depends on where the force acts, whereas a force does not.

9 Angular momentum, kinetic energy and $K = L^2/2I$

Since $L = I\omega$, the kinetic energy can be written $K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$. When L is conserved and I changes, K changes inversely with I . In the stool example of Contact 3, I falls from 5.44 to 2.74 kg m² and K rises by the factor 1.99, from 68.7 to 136 J; the person supplies the difference by pulling the masses inwards. When two bodies join and turn together (the merry-go-round, the discs of Tutorial B8), L is conserved but K falls, by the factor $I_1/(I_1 + I_2)$ if one body starts at rest.

10 Extension: Kepler's second law from conservation of angular momentum

A planet of mass m moves under the gravitational force of the Sun, which is directed along the line joining them. About the Sun this force has no torque, so $L = mvr \sin \varphi = mr^2\dot{\vartheta}$ is constant. In a short time dt the line joining the Sun and the planet sweeps an area $dA = \frac{1}{2}r(r d\vartheta)$, so

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\vartheta} = \frac{L}{2m} = \text{constant}.$$

Equal areas are swept in equal times: Kepler's second law. The first and third laws, and the treatment of orbits, belong to the Gravitation lecture.

11 What is not covered

Rotation about a moving axis other than rolling, rotation about an axis that changes direction (precession and gyroscopes), and the vector form of angular momentum for bodies that are not symmetric about the axis (Young and Freedman, §10.7) are beyond this course.